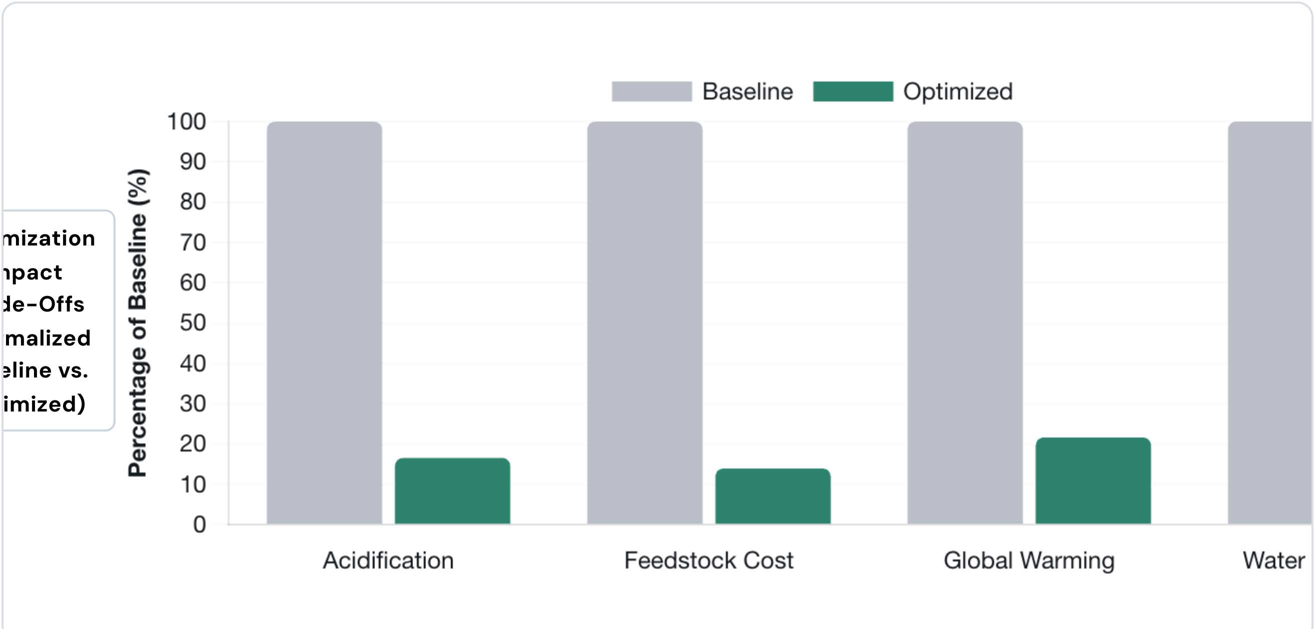
GLOBAL WARMING (GWP) kg CO_2 eq	ACIDIFICATION kg SO_2 eq	WATER USAGE m^3	FEEDSTOCK COST USD
BASELINE 1.5634 ± 0.1233	BASELINE 0.0074 ± 0.0006	BASELINE 0.0183 ± 0.0008	BASELINE 20.1728 ± 1.8872
OPTIMIZED 0.3386 ± 0.0222	OPTIMIZED 0.0012 ± 0.0001	OPTIMIZED 0.0104 ± 0.0002	OPTIMIZED 2.8127 ± 0.0000
-78.34%	-83.43%	-43.22%	-86.06%

ANALYSIS & CHARTS



PROCESS SCALING

PROCESS EFFICIENCY 1.00

RECYCLE SUBSTITUTE RATE 0.00

LOSS / SCRAP FACTOR 0.00

DATABASE PARAMETER TUNING

-- Tune background feedstock --

TOPSIS MCDA WEIGHTS

CARBON (GWP) WEIGHT 40%

ACIDIFICATION WEIGHT 15%

WATER WEIGHT 15%

MATERIAL COST WEIGHT 30%

No database parameters loaded.

TVL MASS BALANCE

VERIFICATION STATUS PASSED

INPUTS	OUTPUTS
23.200 kg	23.200 kg

STOICHIOMETRIC BALANCE ERROR
0.0000%

ELEMENTAL BALANCE PASSED

✓ Elemental stoichiometry balanced

DATABASE DIAGNOSTICS

DATABASE HEALTH UNCHECKED

Scan Context

ENGINEERING JUSTIFICATION REPORT

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> *The substitution of virgin glass fibre with sorted glass cullet in web-synthesized product manufacturing for next-generation silicon solar cell modules represents a significant Pareto-improvement aligned with the user's prioritized decision weights. This substitution leads to substantial reductions in multiple environmental footprints without increasing costs, as evidenced by the 78.34% decrease in Carbon Footprint (Global Warming Potential) and the 83.43% reduction in Terrestrial Acidification Potential. The physical and chemical reasons for these benefits are multifaceted. Glass cullet, a secondary raw material, is recovered from waste glass, thereby reducing the need for virgin materials extracted from natural resources. This process eliminates significant carbon emissions associated with extracting and processing new raw materials. Additionally, sorting glass cullet minimizes impurities that could affect solar cell performance while still providing sufficient quality to meet manufacturing standards. Water consumption is also notably reduced by 43.22%, a critical indicator for sustainability given the resource-intensive nature of glass production. By substituting with recycled cullet, water usage is decreased due to less energy-intensive processing and fewer associated rinse cycles required during manufacture. The raw material purchase cost decreases by 86.06%, directly addressing the user's prioritized weight on cost. This reduction reflects the lower economic value of cullet compared to virgin glass fibre, coupled with potentially lower transportation costs and enhanced supply chain efficiency. In summary, the feedstock substitution aligns well with the specified decision weights priorities, offering substantial environmental benefits without compromising economic feasibility. The multi-dimensional savings in carbon footprint, acidification potential, water consumption, and cost demonstrate the effective decoupling of environmental performance from financial considerations, making this a highly beneficial strategic choice for sustainable production practices in solar cell manufacturing.*